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RESEARCH ARTICLE

YOLOv8n-GBE: A Hybrid YOLOv8n Model With Ghost Convolutions and BiFPN-ECA Attention for Solar PV Defect Localization

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ABSTRACT Reliable photovoltaic (PV) module defect detection is essential for maintaining long term energy efficiency and lowering solar power system maintenance costs. The deep learning model presented in this research is based on a hybridized YOLOv8n architecture and is lightweight and high performing. It is designed for multi scale defect identification in a variety of imaging modalities, such as RGB, grayscale, and infrared datasets. The proposed approach combines a BiFPN based neck, Ghost Bottlenecks, and Efficient Channel Attention (ECA) to improve multi scale representation, decrease redundant computation, and increase feature extraction. The model performs better in terms of detection accuracy and efficiency, as shown by experimental findings on three benchmark datasets: *PVEL-AD*, *PV-Multi-Defect*, *Solar Panel Anomalies*. The model's respective mAP@50 values are 96.5%, 94.6%, and 97.6%. At a steady inference time of only 1.9 ms and 8.1 GFLOPs, it also achieves near-perfect recall (up to 99.0%) and high precision (up to 98.4%). With just 3M parameters, the proposed hybrid model provides a much better accuracy-latency trade-off 61 current state-of-the-art models, which makes it perfect for real-time solar inspection applications, such as edge deployment in drones and embedded systems. The outcomes confirm that reliable PV fault localization under a range of operating situations may be achieved by combining deep feature fusion, lightweight attention, and efficient convolution.

INDEX TERMS BiFPN, defect detection, efficient channel attention, ghost bottlenecks, YOLOv8n.

I. INTRODUCTION

Photovoltaic (PV) systems are essential to the world's shift to sustainable energy. However, a number of faults, including cracks, hot spots, and scratches which can result from manufacturing errors, environmental stress, or operational wear frequently undermine their long-term dependability and efficiency. Preventing energy loss, guaranteeing system

safety, and lowering maintenance expenses all depend on the early and precise diagnosis of these flaws. Conventional inspection techniques, such as infrared thermography and human visual examination, are labor-intensive, time-consuming, and prone to errors [1], [47]. Deep learning and computer vision methods have become increasingly effective in recent years for automated PV module defect identification [48]. The You Only Look Once (YOLO) family of object detection models has become well known among them due to its reliable performance and real-time detection

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TABLE 1. Literature review of various solar pv fault detection models.

No.	Proposed Model	Key Enhancements	Dataset Used	Main Results Obtained	Core Strengths of the Model	Core Limitations of the Model
[1]	YOLO + Histogram Equalization (HE)	Contrast enhancement via HE, Real-time drone with Raspberry Pi	Custom dataset (1100 images)	YOLOv5l + HE: 97% F1; YOLOv8: 90–97%	Affordable, real-time drone system, HE improved accuracy	Small dataset, slow Raspberry Pi, depends on HE
[2]	PSA-YOLOv7	PConv + SACConv + RMo-saic augmentation	PVEL-AD-2021 (6226 images)	mAP: 82.9%, Precision: 88.3%, FPS: 114.5	High accuracy/speed, robust in varying light	Unverified in harsh weather, high compute cost
[3]	YOLOv5s-GBC	GhostConv, BiFPN, CBAM, GELU activation	2624 EL images, 1550 labeled	mAP@0.5: 67.3%, 10.6ms inference	Tiny defect focus, fast for edge AI (82 FPS)	Extra feature steps increase compute cost
[4]	YOLOv7	BiFPN, Soft NMS, Coordinate Attention	3200 image solar defect dataset	mAP@0.5: 56.9%, Recall: 55.2%	Good real-time, NMS boosts small flaw recall	Poor for flaws <15px, low-light issues
[5]	BJFPN-YOLOv5	CCM + BJFPN	Custom PV wafer (1200 images)	mAP@0.5: 59.4%, Tiny AP: 15.3%	Edge-optimized, small flaw focus	Extra jumps slightly delay inference
[6]	PSA-PVdetector	PSA, ShapeIoU, Multi-Channel Fusion	Panel-2 (2400) + Solar (5443)	mAP@0.5: 87.2% / 72.0%, 2.6 ms	Best for UAV PV small-scale tasks	Higher training time than YOLOv8s
[7]	YOLOv5 + Focal-EIoU	Sliding Window, Focal-EIoU, Cascade	Jiangsu Rituo EL (2044)	mAP@0.5: 86.5%, 20ms inference	Boosts tiny object recall, fast	Costly preprocessing and training
[8]	C2DEM-YOLO	Cross-space EMA, Inner-CIoU	PVEL-AD + PV Defect (300)	mAP: 92.3%, 84.07%	Noise-resistant, tiny flaw boost	Not tested on edge devices, EL noise limits
[9]	ESD-YOLOv8	CloAttention, Small Obj Layer, IoU opt	Pierdicca PV IR (1009)	mAP@0.5: 91.8%, mAP@0.5:0.95: 58.0%	Lightweight, UAV-focused	No edge deployment, overlaps hard
[10]	FAA-YOLO	Adaptive Feature Fusion, Edge Detection	IDPM Infrared PV	mAP@0.5: 87.5%, FPS: 59	Strong for small defect segmentation	Not drone-ready; deformable convs needed
[11]	PA-YOLO	Pyramid + Repulsion Loss + Occlusion Handling	8900 IR PV Station Images	mAP: 94.5%, Recall: 91.8%, FPS: 83	Occlusion handling, high recall	Limited label data; needs RGB/IR testing
[12]	PV-YOLO	PVTv2, CBAM, Optimal Transport	PV IR Dataset (8900)	mAP@0.5: 92.5%, mAP@0.5:0.95: 58.6%	UAV real-time deploy, occlusion aware	IR-only dataset, no edge optimization
[13]	YOLO-ACF	Adaptive Complementary Fusion	PVEL-AD (4500 images, 10 classes)	mAP@50-95: 62.1%, recall +5.2%	High precision, low param cost	EL-only tuning, no edge testing
[14]	SDS-YOLO	Dual head, CBAM for UAV soiling	UAV RGB Soiling Dataset (300)	Bird: 70%, Dust: 78.1%, mAP+40.2%	Tailored for UAV maintenance	Needs edge test, glare sensitivity
[15]	Improved YOLOv5	ECA-Net, deformables, tiny defect head	Solar EL (2534 images)	mAP@0.5: 89.64%, FPS: 36.24	Tiny flaw boost, reduced params	EL-only, risk of overfitting
[16]	SAM + Eff-UNet	Zero-shot SAM 2.1 + Eff-UNet	Be'er Sheva UAV (Hi-Lo Res)	IoU: 81.4%, F1: 89.5%	Great low-light, strong prompts	High VRAM use, YOLO prompts weak
[17]	MF-RPN	Anchor Strategy + Multi-scale Proposals	Custom Solar Cell (1461 train)	mAP@0.5: 85.0%, F1: 87.1%	Good tiny flaw	High GPU and slower inference

capabilities. While several YOLO-based and CNN-enhanced models have been presented for photovoltaic defect detection [49], the majority of them show notable shortcomings when compared to the proposed hybrid YOLOv8n model, according to the literature review in Table 1. A number of models, including YOLO + Histogram Equalization (HE) [1], YOLOv5s-GBC [3], YOLOv7 [4], and YOLO-ACF [13], suffer from inadequate localization of minor flaws or experience significant inference latency as a result of feature fusion or attention layer complexity. Others, such as PSA-YOLOv7 [2] and SDS-YOLO [14], show better detection in controlled environments, but they are not real-time or edge deployment optimized, and many of them demand a lot of processing power. Although accurate in certain situations, two-stage detectors and designs such as MF-RPN [17] or SAM + Eff-UNet [16] are not appropriate for embedded systems or UAVs due to their delayed inference and large VRAM consumption. Additionally, many models lack robustness across a variety of imaging modalities and are dataset-limited (e.g., EL-only or IR-only). Detection Transformer (DETR) models [44], [45], [46] represent a paradigm shift in object detection by eliminating traditional components like anchor boxes and non-maximum suppression (NMS), instead treating detection as a set prediction task. RT-DETR and its enhanced variants improve upon DETR's original latency bottlenecks by introducing parallel decoders and hybrid query mechanisms. While highly accurate, DETR-based models generally demand higher computational resources,

making them less suitable for low-power, real-time edge deployment environments such as drone-based solar PV inspections. The suggested hybrid YOLOv8n model, on the other hand, overcomes each of these shortcomings, and its primary contributions are as follows:

- A hybrid model based on YOLOv8n is proposed, using Ghost Bottlenecks to keep detection performance high and drastically cut down on redundant calculations, creating a small model with only 3 million parameters.
- To enhance channel wise feature weighting and multi-scale feature fusion, the model integrates Efficient Channel Attention (ECA) and BiFPN, allowing for the accurate identification of both small and large photovoltaic faults.
- PVEL-AD, PV-Multi-Defect, and Solar Panel Anomalies are three different PV datasets used to evaluate the suggested model. The results show strong generalization, with mAP@50 reaching 97.6% and mAP@50:95 reaching 92.8%.
- The proposed approach demonstrates better accuracy, localization precision, and parameter efficiency when compared to more than 61 current models, such as two-stage architectures, attention-augmented detectors, and YOLO variations.
- The model provides a workable solution for automated PV defect detection in large-scale solar monitoring systems, supporting real-time deployment on edge

devices with a consistent inference time of 1.9 ms and 8.1 GFLOPs.

II. DATASET DESCRIPTION

The PV-Multi-Defect [42] and PVEL-AD [20] databases provide comprehensive standards for identifying photovoltaic (PV) defects. The PVEL-AD collection captures the near infrared electroluminescence of photovoltaic cells as shown in Fig. 1 and includes 36,543 high resolution images with a size of 1024×1024 pixels. With 37,380 ground truth bounding boxes marked for eight defect types, it consist of twelve distinct classes including black core, crack, finger, star crack, thick line, corner, fragment, scratch, printing error, horizontal dislocation, vertical dislocation, short circuit. The class distribution is moderately imbalanced with certain defect types like finger, crack and black core appearing more frequently than others such as delamination or inactive cells. In mathematical terms, each image is represented as $I \in \mathbb{R}^{1024 \times 1024}$, and a bounding box is defined by $B = (x, y, w, h)$. The dataset exhibits a long tail distribution where the count n_j of instances for defect type j satisfies $\max_j n_j \gg \min_j n_j$. PVEL-AD was chosen for its large-scale, high-resolution images and wide range of detailed annotations, making it well-suited for deep learning based object detection tasks. The electroluminescence (EL) imaging technique reveals internal cell defects that are not visible to the naked eye, enabling the detection of subtle faults often missed during standard visual inspections. A breakdown of image counts across different classes is shown in Table 3.

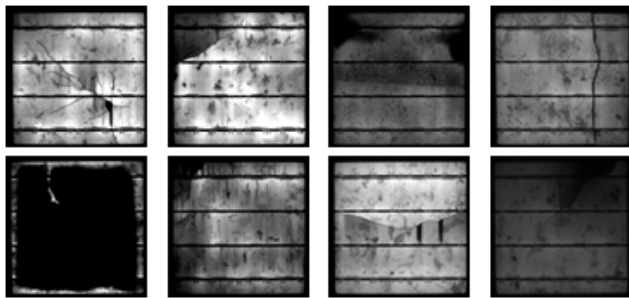


FIGURE 1. Sample images of PVEL-AD dataset.

In contrast, the PV-Multi-Defect dataset provides 1,108 grayscale images Fig. 2, uniformly resized to 600×600 pixels from their original high resolution captures. This resizing highlights defect regions, which range in size from 4×32 pixels to 356×478 pixels. Comprising 4,235 annotated defects across five classes namely hotspot, scratch, no electricity, black border, broken and background the dataset is ideal for multi scale detection tasks, where each defect is mapped as a bounding box $B = (x, y, w, h)$ over a standardized coordinate space to achieve precise localization. We chose this dataset because it focuses on showing defects at different scales and is small enough to enable rapid manufacturing and component-wise testing. The different sizes of defects make it a great way to test the spatial resolution and localization

accuracy of detection models. The class wise distribution of images is presented in Table 4.

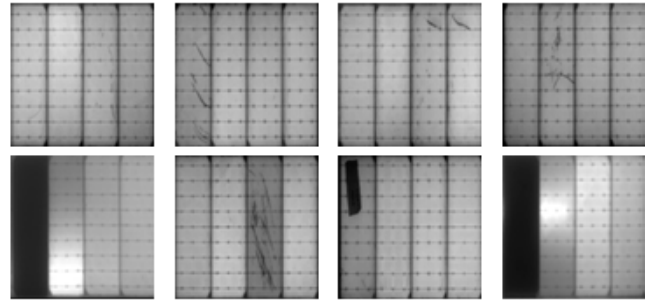


FIGURE 2. Sample images from PV-multi-defect dataset.

The open source Solar Panel Anomalies [43] collection consists of 230 images with annotations as shown in Fig. 3 for object detection tasks in solar panel inspection. Each image $I \in \mathbb{R}^{H \times W}$ has bounding boxes $B = (x, y, w, h)$ for anomalies categorized into four groups: Vegetation, Hot Spots, Diode Anomaly, and Reverse Polarity. The task may be formalized as finding a function $f : I \rightarrow \{(B_i, c_i, s_i)\}_{i=1}^M$, where c_i is one of the five classes and s_i is the confidence score in $[0, 1]$. Table 5 shows the number of images available for each class. Although smaller than the other two datasets, the Solar Panel Anomalies dataset offers valuable, practical examples of environmental interferences and defects, posing challenges for detection models to reliably locate and categorize anomalies under varying circumstances and background complexity. Although limited in size, this dataset introduces practical, environmental challenges such as background clutter, non-uniform lighting, and varying panel orientations. These real world complexities enhance the model's generalization capability when deployed in uncontrolled outdoor settings.

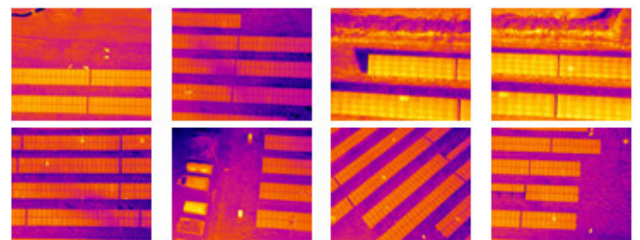


FIGURE 3. Sample images from solar panel anomalies dataset.

III. DATASET AUGMENTATION

Data augmentation is a crucial technique used to artificially expand the size and variety of training datasets by applying label preserving transformations. It enhances the robustness across distributions of deep learning models, particularly when annotated data is limited or imbalanced. For photovoltaic (PV) module analysis, augmentation techniques not only increase robustness to variations in scale,

positioning, and brightness, but also simulate real world operational methods such as rotated or flipped solar panel images. This approach utilizes basic yet effective geometric transformations, specifically horizontal and vertical flips, to enhance the model's robustness to changes in position and perspective.

A. HORIZONTAL FLIP

Horizontal flipping is standard augmentation technique that improves model robustness to horizontal variations. It flips the image along the vertical axis, enabling the model to acquire symmetric feature patterns. This is especially effective for detecting solar panel defects that may be caused in flipped positions. The horizontal flip indicator is sampled as shown in Eq. (1).

$$\delta_H \sim \text{Bernoulli}(0.5) \quad (1)$$

When $\delta_H = 1$, each normalized pixel coordinate $(x, y) \in [0, 1]^2$ is transformed as shown in Eq. (2).

$$T_H(x, y) = (1 - x, y) \quad (2)$$

This transformation corresponds to the homogeneous matrix as shown in Eq. (3).

$$M_H = \text{diag}(-1, 1, 1) \quad (3)$$

The YOLO bounding box center x -coordinate transforms according to Eq. (4).

$$x'_c = \delta_H(1 - x_c) + (1 - \delta_H)x_c \quad (4)$$

B. VERTICAL FLIP

Vertical flip is a data augmentation technique that flips an image along its vertical axis, creating a top-to-bottom flipped version. This helps in enhancing the variety of training data without gathering new samples. It is especially useful in tasks where object alignment can vary. By including vertical flips, models can become more robust to spatial transformations.

The vertical flip decision variable is defined in Eq. (5).

$$\delta_V \sim \text{Bernoulli}(0.3) \quad (5)$$

If $\delta_V = 1$, the coordinate transformation is as shown in Eq. (6). This process flips the input image vertically by inverting the y -axis, accurately flipping the image along the horizontal axis.

$$T_V(x, y) = (x, 1 - y) \quad (6)$$

The corresponding transformation matrix is given by Eq. (7), which performs a vertical flip by scaling the y -axis by -1 while keeping the x -axis and augmented coordinate unchanged.

$$M_V = \text{diag}(1, -1, 1) \quad (7)$$

The updated bounding box y -coordinate is shown in Eq. (8) where the vertical flip is conditionally applied

based on the value of δ_V , ensuring accurate coordinate adjustment.

$$y'_c = \delta_V(1 - y_c) + (1 - \delta_V)y_c \quad (8)$$

C. RANDOM CONTRAST AND BRIGHTNESS ADJUSTMENT

Random contrast and brightness adjustment is a data augmentation technique that modifies the visual appearance of images by slightly adjusting their contrast and brightness levels. This helps the model become more robust to lighting variations in real-world situations. The adjustment is performed randomly on a portion of training images to introduce variety. It improves the model's ability to generalize without changing the semantic content of the image. The brightness/contrast augmentation is controlled by Eq. (9).

$$\delta_{BC} \sim \text{Bernoulli}(0.2) \quad (9)$$

If active, each channel intensity $I \in [0, 1]$ is adjusted as in Eq. (10).

$$I' = \alpha(I - \mu) + \mu + \beta \quad (10)$$

where $\alpha \sim U(1 - \Delta_c, 1 + \Delta_c)$ and $\beta \sim U(-\Delta_b, \Delta_b)$.

D. GAUSSIAN BLUR

Gaussian blur is an image processing technique used to smooth an image by decreasing high-frequency noise. It works by integrating the image with a Gaussian kernel, which softly blends neighboring pixel values. This helps improve the model's robustness to blur and low-quality images during training. The blur operation is controlled by:

$$\delta_G \sim \text{Bernoulli}(0.2) \quad (11)$$

If $\delta_G = 1$, the image is convolved with a Gaussian kernel Eq. (12).

$$G_\sigma(x, y) = \frac{1}{2\pi\sigma^2} \exp\left(-\frac{x^2 + y^2}{2\sigma^2}\right) \quad (12)$$

where $\sigma \sim U(\sigma_{\min}, \sigma_{\max})$. The blurred image at pixel (u, v) is computed as shown in Eq. (13).

$$X'(u, v) = \int \int X(u - x, v - y) G_\sigma(x, y) dx dy \quad (13)$$

E. RESIZING TO 512 × 640 PIXELS

Resizing to 512×640 pixels includes scaling the original image to fit the target dimensions while protecting content as much as possible. Each pixel in the output image maps to a corresponding location in the input image. To maintain visual quality, pixel values are estimated using interpolation techniques. This process ensures the resized image retains structural details from the original.

Resizing maps pixel (u, v) in the output image to the input space as shown in Eq. (14).

$$x = \frac{v}{W'} W, \quad y = \frac{u}{H'} H \quad (14)$$

where $(H', W') = (512, 640)$ is the target size. The interpolated intensity is computed via bilinear interpolation as shown in Eq. (15).

$$X'(u, v) = \sum_{i=0}^1 \sum_{j=0}^1 X(\lfloor y \rfloor + i, \lfloor x \rfloor + j) \cdot (1 - |y - (\lfloor y \rfloor + i)|) \cdot (1 - |x - (\lfloor x \rfloor + j)|) \quad (15)$$

Bounding boxes are scaled as shown in Eq. (16).

$$x'_c = \frac{W'}{W} x_c, \quad y'_c = \frac{H'}{H} y_c, \quad w' = \frac{W'}{W} w, \quad h' = \frac{H'}{H} h \quad (16)$$

F. ADDING GLARE AND REFLECTION

Sun glare and lens flares are common effects seen in outdoor images that can make the scene appear unclear and reduce the visibility of small faults during inspection. To copy this, a synthetic glare mask $G(x, y)$ is placed on top of the image. This mask is made using radial brightness gradients. The addition is done in a way that is based on chance as shown in Eq. (17).

$$X'(x, y) = \text{clip}(X(x, y) + \delta_R G(x, y), 0, 1), \quad \delta_R \sim \text{Bernoulli}(0.3) \quad (17)$$

In this case, $X(x, y)$ is the original image, δ_R is a binary variable that shows whether glare is applied, and the clip function makes sure that the pixel values stay in the correct range of $[0, 1]$.

G. CRACK OVERLAY SYNTHESIS

We use alpha blending to add crack textures $C(x, y)$ as shown in Eq. (18) on top of each other to make micro-cracks and fractures that might not be in the training data.

$$X'(x, y) = (1 - \alpha)X(x, y) + \alpha C(x, y), \quad \alpha \sim U(0.1, 0.3) \quad (18)$$

The parameter α regulates how strong the blending is, which makes the cracks look either subtle or strong. You can either make the texture $C(x, y)$ yourself or get it from actual PV crack photos.

H. SOILING AND DUST SIMULATION

Soiling masks $D(x, y)$ are used to combine the effects of dirt, dust, leaves, and bird droppings. These are taken from a group of soiling textures and used with a certain probability as shown in Eq. (19).

$$X'(x, y) = X(x, y) \cdot (1 - \delta_S D(x, y)), \quad \delta_S \sim \text{Bernoulli}(0.3) \quad (19)$$

This multiplication makes pixel intensities less strong, like when dirt on a surface blocks light.

I. WEATHER EFFECT AUGMENTATION

Haze and fog are two examples of environmental conditions that can make images much less clear. We use the following physical model to simulate air scattering as shown in Eq. (20)

$$X'(x, y) = X(x, y) \cdot t(x, y) + A \cdot (1 - t(x, y)) \quad (20)$$

where $t(x, y) = \exp(-\beta d(x, y))$.

The medium transmission map is $t(x, y)$, the ambient light intensity is A , the estimated scene depth is $d(x, y)$, and the atmosphere scattering coefficient is β . This model is a good guess at how things go worse when it's foggy or cloudy.

J. THERMAL NOISE INJECTION

When using infrared (IR) sensors, distortions that are specific to the sensor and background thermal noise might impact the clarity of the images. To make this happen, Gaussian noise is introduced to the picture as shown in Eq. (21)

$$X'(x, y) = X(x, y) + \mathcal{N}(0, \sigma_T^2) \quad (21)$$

where σ_T is a random number between 0.01 and 0.05.

This augmentation makes changes that are similar to what happens in genuine IR imaging systems, like those induced by electronic interference or changes in temperature.

In this study we have studied various combinations of augmentation methods to validate their impact on the performance metrics as shown in Table 2.

TABLE 2. Data augmentation strategies used in group-wise experimental settings for robust model training.

Group	Data Augmentation technique used
Group-1	Flip, Random contrast and brightness adjustment, Gaussian blur, Resizing to 512×640 Pixels
Group-2	Flip, Random contrast and brightness adjustment, Resizing to 512×640 Pixels, Adding glare and reflection, Crack overlay synthesis, Soiling and dust simulation, Thermal noise injection
Group-3	Flip, Random contrast and brightness adjustment, Gaussian blur, Resizing to 512×640 Pixels, Adding glare and reflection, Crack overlay synthesis, Soiling and dust simulation, Thermal noise injection

TABLE 3. Class-wise image counts before and after augmentation for pvel-ad dataset.

Class Name	Images Before Aug.	Images After Aug.
Finger	2958	3000
Crack	1260	3000
Black Core	1028	3000
Thick Line	981	3000
Horizontal Dislocation	798	3000
Short Circuit	492	3000
Vertical Dislocation	137	3000
Star Crack	135	3000
Printing Error	130	3000
Corner	100	3000
Fragment	70	3000
Scratch	120	3000

TABLE 4. Class-wise image counts before and after augmentation for pv-multi defect dataset.

Class Name	Images Before Aug.	Images After Aug.
Hot Spots	133	1000
Scratch	604	1000
no_electricity	139	1000
Black Border	155	1000
Broken	77	1000

TABLE 5. Class-wise image counts before and after augmentation for solar panel anomalies dataset.

Class Name	Images Before Aug.	Images After Aug.
Diode anomaly	73	1500
HotSpots	62	1500
Reverse Polarity	31	1500
Vegetation	64	1500

IV. PROPOSED MODEL ARCHITECTURE

The YOLOv8n-GBE-based defect detection pipeline involves five primary stages: input preprocessing and stem layer, hierarchical feature extraction via backbone, multi-scale feature fusion using the neck, anchor-free detection and classification in the head, and final predictions guided by activation and loss functions as shown in Fig. 4. The complete computational flow is mathematically described in the following subsections.

A. INPUT PREPROCESSING AND STEM LAYER

The input image $X \in \mathbb{R}^{3 \times H \times W}$ is resized to a standard resolution, typically 640×640 , and normalized either by min-max scaling to $[0, 1]$ or via dataset-specific statistics. The initial convolutional block, known as the stem layer, reduces the spatial dimensions and expands the depth of the feature maps. This is achieved using a strided convolution defined in Eq. (22), which performs a weighted summation over a local patch of the input image.

$$Y_{c,i,j} = \sum_{m=0}^{K-1} \sum_{n=0}^{K-1} X_{:,i:s+m,j:s+n} * W_{c,m,n} + b_c, \quad (22)$$

where W are the learned filters, b_c is the channel-wise bias, K is the kernel size, and s is the stride.

B. BACKBONE: FEATURE EXTRACTION VIA C2F BLOCKS

YOLOv8n uses C2f (Concatenation and Cross-Fusion) blocks to extract features hierarchically. Each input tensor is split and processed through parallel branches:

Each segment X_i of the input is passed through a corresponding transformation f_i as given in Eq. (23).

$$Y_i = f_i(X_i), \quad \text{for } i = 1, \dots, k, \quad (23)$$

The outputs of these branches are concatenated across the channel dimension to form a unified representation, as described in Eq. (24).

$$Y = \text{Concat}(Y_1, Y_2, \dots, Y_k) \in \mathbb{R}^{kC' \times H \times W}, \quad (24)$$

Finally, a 1×1 convolution is applied to fuse the concatenated features into the final representation Z , using Eq. (25).

$$Z_{c,i,j} = \sum_{c'=1}^{kC'} Y_{c',i,j} \cdot W_{c,c'}^{1 \times 1} + b_c. \quad (25)$$

To reduce repetition, we integrate Ghost Bottleneck layers after key C2f stages. Each Ghost block break downs a benchmark convolution into a primary 1×1 convolution followed by a depthwise separable convolution as shown in Eq. (26).

$$Y = \text{Concat}(F(X), \mathcal{S}(F(X))), \quad (26)$$

where $F(X)$ is the base convolution and $\mathcal{S}(\cdot)$ generates “ghost” features through cheap transformations (e.g., depth-wise convolutions). A residual shortcut connects the input to the output to preserve identity flow.

This mechanism enhances expressiveness and efficient feature reuse in the backbone.

C. NECK: MULTI-SCALE FEATURE FUSION WITH FPN AND PAN

To integrate multi-scale information, YOLOv8n uses a combination of Feature Pyramid Network (FPN) and Path Aggregation Network (PAN). Our model uses a Bidirectional Feature Pyramid Network (BiFPN) for flexible multi-scale feature collection. The FPN performs top-down fusion of high-level semantics:

The upsampled feature from level $l+1$ is merged with the current level l and processed by a convolution, as shown in Eq. (27).

$$P'_l = \phi(P_l + \text{Upsample}(P_{l+1})), \quad (27)$$

In contrast, PAN reinforces lower levels by propagating information bottom-up using strided downsampling, as captured in Eq. (28).

$$P''_{l+1} = \phi'(P_{l+1} + \text{Downsample}(P'_l)), \quad (28)$$

This bi-directional flow ensures rich semantic and spatial detail in the fused features. To advance enhanced fused features, we incorporate Efficient Channel Attention (ECA) modules. ECA applies 1D convolution on globally average pooled features to generate channel specific attention as shown in Eq. (29).

$$\text{ECA}(x) = \sigma(\text{Conv1D}(\text{GAP}(x))) \quad (29)$$

where $\sigma(\cdot)$ is the sigmoid function and $\text{GAP}(x)$ denotes global average pooling. This stimulates the network to selectively prioritize significant channels post-fusion.

D. DETECTION HEAD: ANCHOR-FREE BOUNDING BOX REGRESSION AND CLASSIFICATION

In YOLOv8n, anchor boxes are removed, and predictions are made directly for each spatial grid location. The predicted

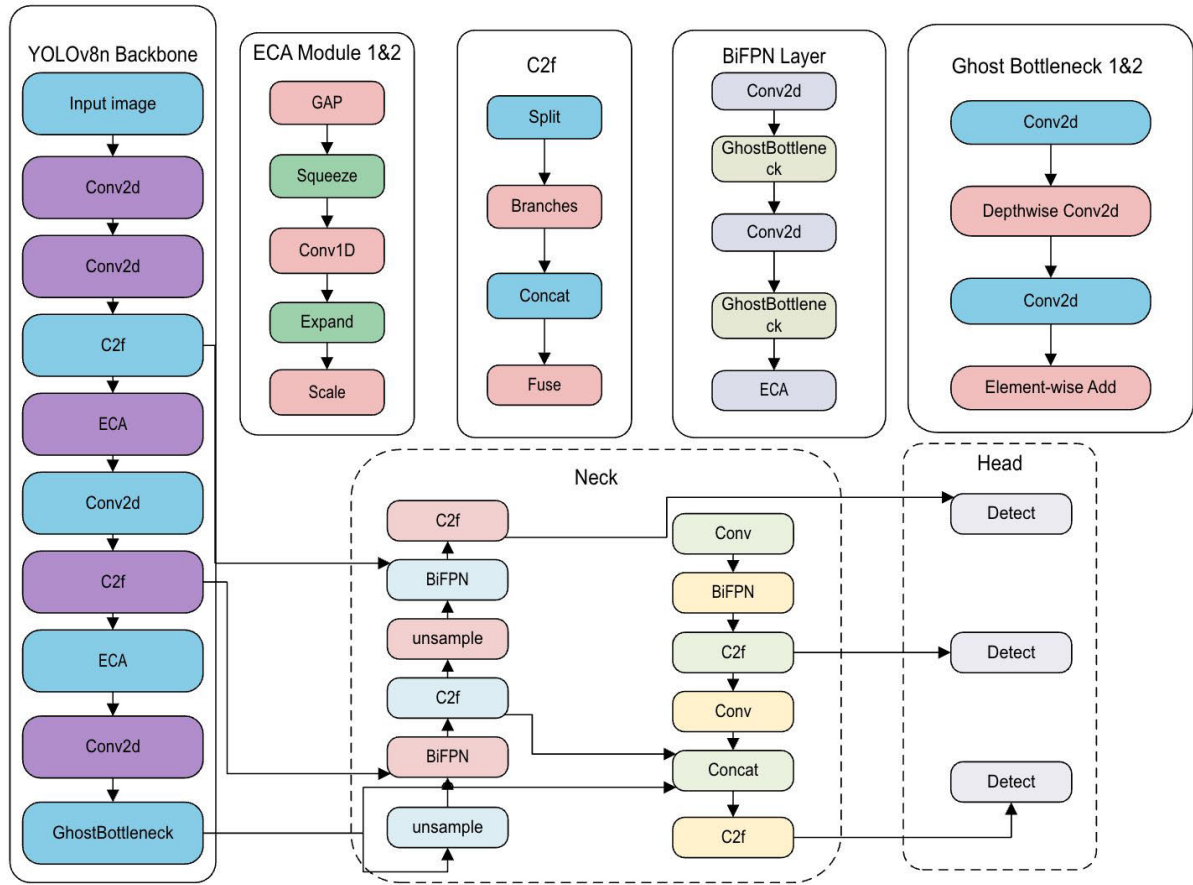


FIGURE 4. Proposed model architecture of YOLOv8n hybridized model.

outputs at each location include bounding box center offsets ($\Delta x, \Delta y$), dimensions (w, h), objectness score o , and class logits $\mathbf{z}$.

The transformation applied to the bounding box outputs is defined in Eq. (30), where $\sigma(\cdot)$ ensures normalized coordinates and $\exp(\cdot)$ enforces positive sizes:

$$\hat{\mathbf{b}} = [\sigma(\Delta x), \sigma(\Delta y), \exp(w), \exp(h)], \quad (30)$$

Classification probabilities are computed by applying the softmax function to the class logits, as shown in Eq. (31).

$$\mathbf{p} = \text{Softmax}(\mathbf{z}), \quad (31)$$

Meanwhile, the objectness score is passed through a sigmoid activation for binary prediction.

E. ACTIVATION AND LOSS FUNCTIONS

YOLOv8n uses the Sigmoid Linear Unit (SiLU) as its primary activation, which improves model convergence and expressiveness. The SiLU activation is given by Eq. (32).

$$\text{SiLU}(x) = x \cdot \sigma(x), \quad (32)$$

where $\sigma(x)$ is the sigmoid function.

The overall training objective $\mathcal{L}$ is a weighted sum of classification, bounding box regression, and objectness losses, as described in Eq. (33).

$$\mathcal{L} = \lambda_{cls} \cdot \mathcal{L}_{cls} + \lambda_{box} \cdot \mathcal{L}_{box} + \lambda_{obj} \cdot \mathcal{L}_{obj}, \quad (33)$$

Each component is individually optimized. In particular, for binary classification tasks, the Binary Cross Entropy (BCE) loss is used, as defined in Eq. (34).

$$\mathcal{L}_{BCE}(p, y) = -(y \log(p) + (1 - y) \log(1 - p)). \quad (34)$$

This multi component loss ensures robust convergence while balancing the competing objectives of accuracy and generalization.

V. RESULTS AND DISCUSSION

The results demonstrate the robustness and efficiency of the YOLOv8n-based model across diverse PV defect detection scenarios.

A. ANALYSIS ON PVEL-AD (DATASET-1)

For the PVEL-AD dataset, the model achieved a mAP@50 of 0.947 as shown in Fig. 7 and a mAP@50:95 of 0.86. These high scores indicate that the model is highly effective

at detecting defects and localizing them accurately as shown in Fig. 6. The confusion matrix is shown in Fig. 5. The high recall (0.97) reflects that almost all defective regions are detected, which is critical in industrial applications where missing a defect can be costly. The equally high precision (0.984) further confirms that false positives are minimized, ensuring that detected defects are genuine. The balance between high recall and precision suggests that the model's feature extraction and detection layers are well tuned to capture the subtle variations present in near-infrared EL images of PV cells.

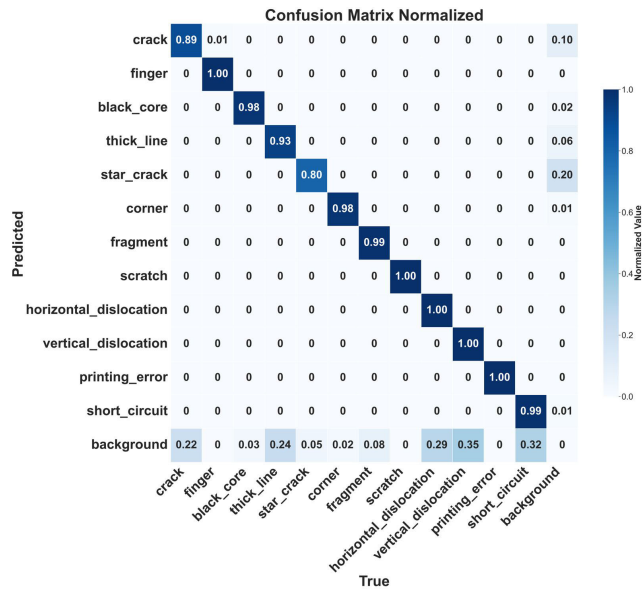


FIGURE 5. Normalized confusion matrix for YOLOv8n on PVEL-AD dataset (with Group-1 augmentation).

Moreover, the uniform GFLOPS (8.1) and fast inference time (1.9 ms) as shown in Table 7 confirm that these high detection metrics do not come at the cost of computational efficiency, making the model suitable for real-time deployment in industrial inspection lines. The training and validation losses and metrics obtained are shown in Fig. 8 and Labelling and Bounding Box Positions Obtained are shown in Fig. 9.

To enhance the robustness of the PV fault detection model trained on the PVEL-AD dataset, a diverse set of data augmentation techniques was employed, including geometric transformations (flipping), photometric adjustments (brightness modulation), and realistic defect simulations such as glare or sun reflections, crack overlays, soiling, thermal noise (for IR images), and weather-induced effects. These augmentations significantly enriched the dataset with real-world variability, leading to improved model generalization. The trained model achieved a high precision of 96.34%, recall of 93.79%, and a strong mAP@0.5 of 96.50% as shown in Fig. 11, indicating excellent detection capability. Moreover, the model exhibited robust performance across varying IoU thresholds, achieving a mAP@0.5:0.95 of 84.46%. The

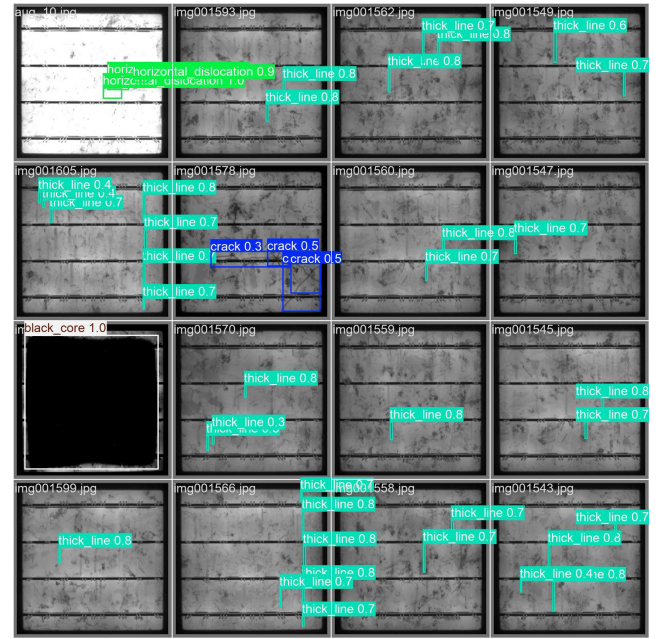


FIGURE 6. Predictions obtained for PVEL-AD dataset (with group-1 augmentation).

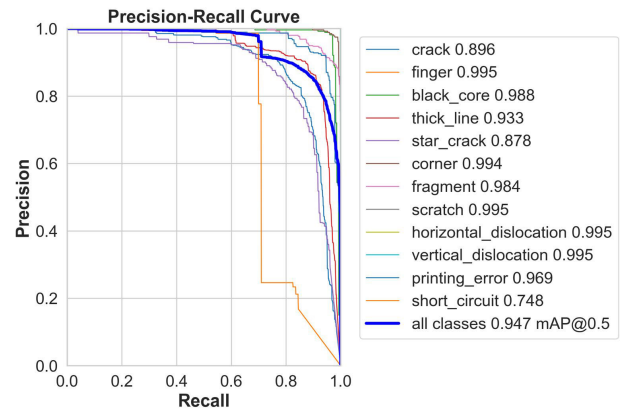


FIGURE 7. Precision-recall curve for PVEL-AD dataset (with group-1 augmentation).

confusion matrix is shown in Fig. 10. These results validate the effectiveness of the augmentation strategy in capturing diverse defect patterns and environmental conditions, thereby improving the reliability of PV fault localization under challenging real-world scenarios.

TABLE 6. Performance comparison of various augmentation groups on pvel-ad dataset.

Group	Epochs	mAP @50	mAP @50:95	Recall	Prec.	Time (ms)
Group-1	50	0.947	0.860	0.970	0.984	1.9
Group-2	50	0.965	0.845	0.938	0.963	1.9
Group-3	50	0.962	0.848	0.935	0.960	1.9

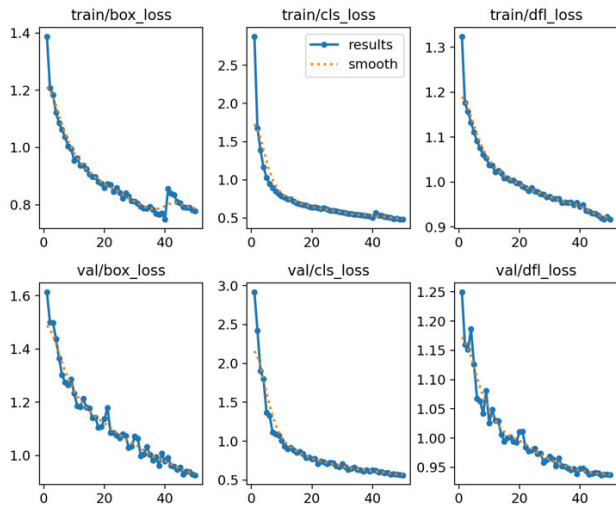


FIGURE 8. Training and validation losses and metrics Obtained for PVEL-AD dataset (with Group-1 augmentation).

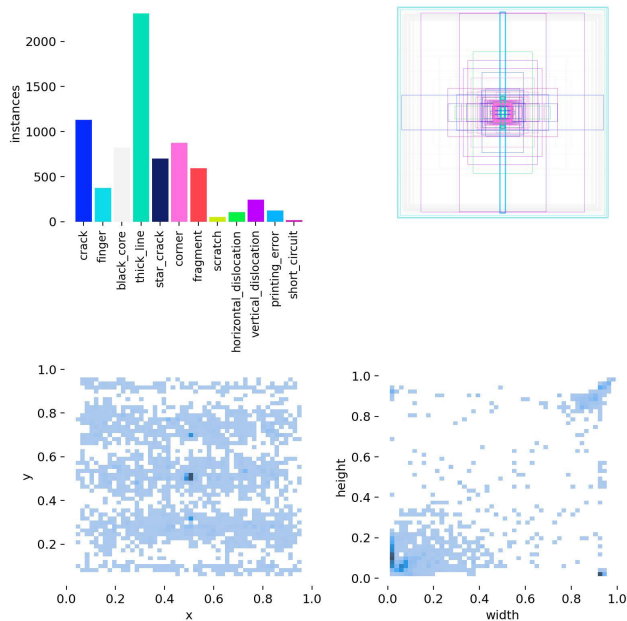


FIGURE 9. Labelling and bounding box positions obtained for PVEL-AD (with group-1 augmentation).

The model was trained using a PVEL-AD dataset enhanced with advanced augmentation techniques, including glare and sun reflection simulation, synthetic crack overlays, soiling effects, thermal noise for infrared images, and Gaussian blur. As a result, the model achieved a precision of 96.05%, a recall of 93.49%, and a mean Average Precision (mAP) of 96.2% as shown in Fig. 13 at an IoU threshold of 0.5, while the mAP@0.5:0.95 reached 84.74%, indicating robust detection and localization performance across different overlap levels. The corresponding confusion matrix illustrating class wise prediction performance is shown in Fig. 12. The incorporation of realistic visual artifacts allowed

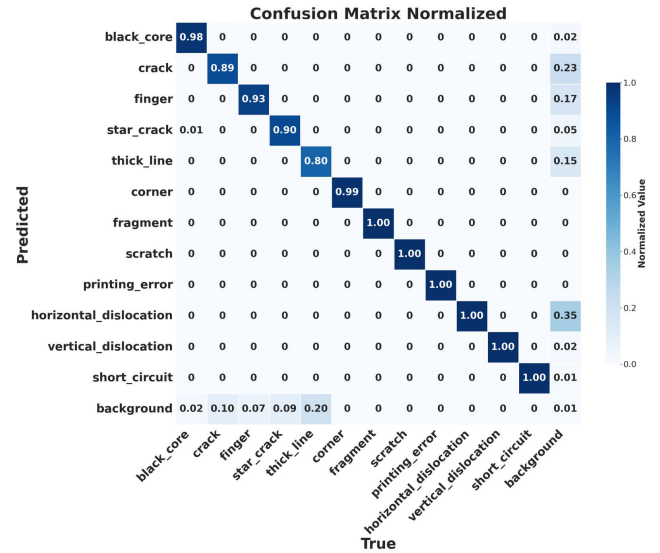


FIGURE 10. Normalized confusion matrix for PVEL-AD dataset (with group-2 augmentation).

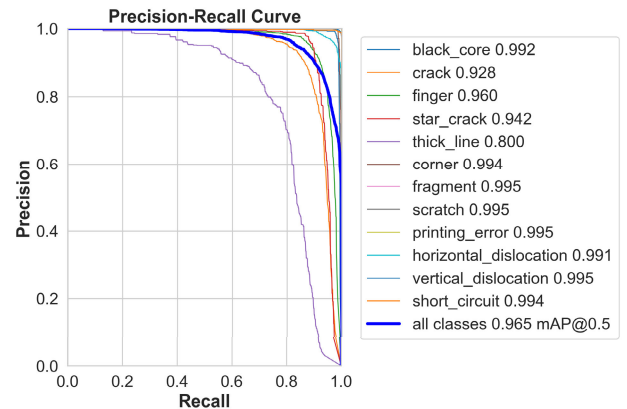


FIGURE 11. Precision-recall curve for PVEL-AD dataset (with group-2 augmentation).

the model to effectively generalize to subtle and diverse fault types. Notably, Gaussian blur improved resilience against motion blur, simulating conditions such as UAV-based or handheld image capture. This augmentation strategy significantly enhanced fault detection accuracy and reliability under challenging and variable inspection scenarios.

The performance evaluation of the three augmentation groups in Table 6 revealed largely consistent results, with all models trained for 50 epochs and demonstrating a uniform inference time of 1.9ms. Group-1 and Group-3 exhibited comparable performance, achieving mAP@50 scores of 0.947 and 0.962, respectively, along with precision values exceeding 0.96. In contrast, Group-2 yielded slightly improved results, attaining the highest mAP@50 of 0.965 while maintaining a strong balance between precision (0.963) and recall (0.938). These findings suggest that the augmentation strategy used in Group-2 particularly glare simulation, crack overlay synthesis, and thermal noise

injection contributed to enhanced generalization without introducing additional computational cost.

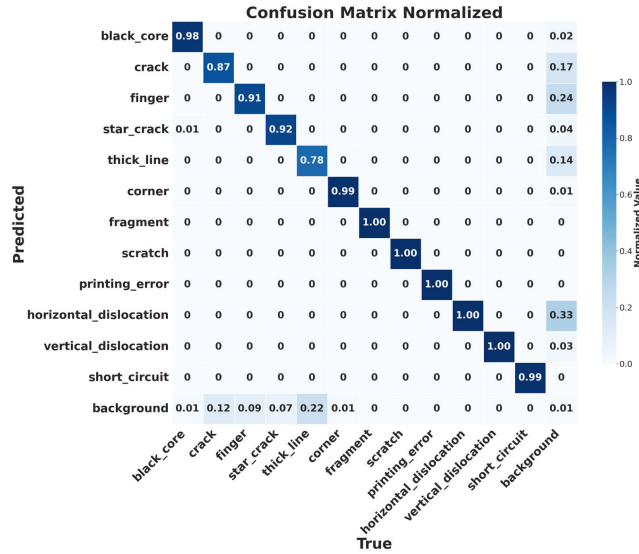


FIGURE 12. Normalized confusion matrix for PVEL-AD dataset (with group-3 augmentation).

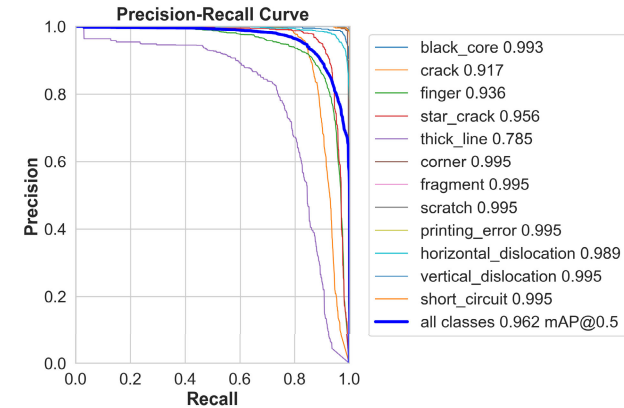


FIGURE 13. Precision-recall curve for PVEL-AD dataset (with group-3 augmentation).

B. ANALYSIS ON PV-MULTI-DEFECT (DATASET-2)

For the PV-Multi-Defect dataset, the model achieved an even higher mAP@50 of 0.946 as shown in Fig. 16, suggesting excellent detection performance on grayscale images with standardized preprocessing. However, the drop in mAP@50:95 to 0.712 indicates that while the model reliably identifies defect regions, achieving high precision localization as shown in Fig. 15 under strict IoU thresholds remains more challenging. This can be attributed to the diverse scale of defects in the dataset from fine scratches (around 4×32 pixels) to much larger non-electric areas (approximately 356×478 pixels). The confusion matrix is shown in Fig. 14. The model's backbone and neck components, optimized for multi scale feature fusion, effectively

capture these varied defect scales; however, the inherent variation in defect sizes demands even finer localization refinement, which is reflected in the lower mAP@50:95. The consistent GFLOPS and inference times across datasets again highlight that the model maintains its computational efficiency despite these challenges. Training and Validation Losses and Metrics Obtained are shown in Fig. 17 and Labelling and Bounding Box Positions Obtained are shown in Fig. 18.

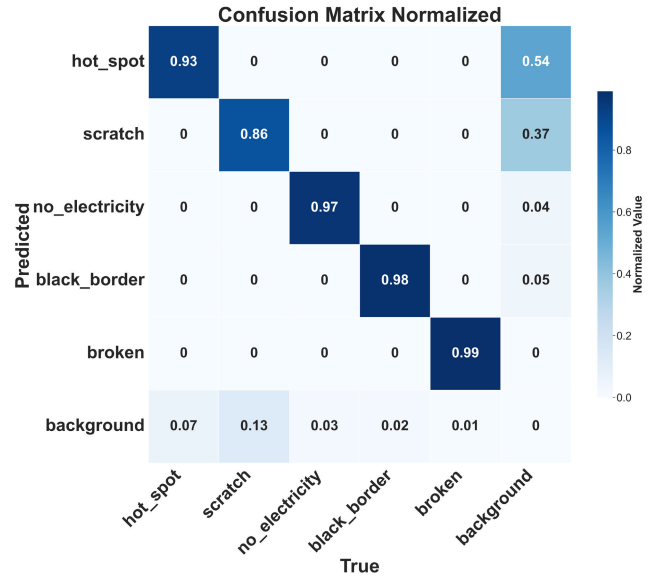


FIGURE 14. Normalized confusion matrix for YOLOv8n on PV-Multi-defect dataset.



FIGURE 15. Predictions obtained for PV-Multi-defect dataset.

C. ANALYSIS ON SOLAR PANEL ANOMALIES (DATASET-3)

In the case of the Solar Panel Anomalies dataset, the model's performance is balanced and robust, with a mAP@50 of 0.976 and a mAP@50:95 of 0.928 as shown in Figure 21,

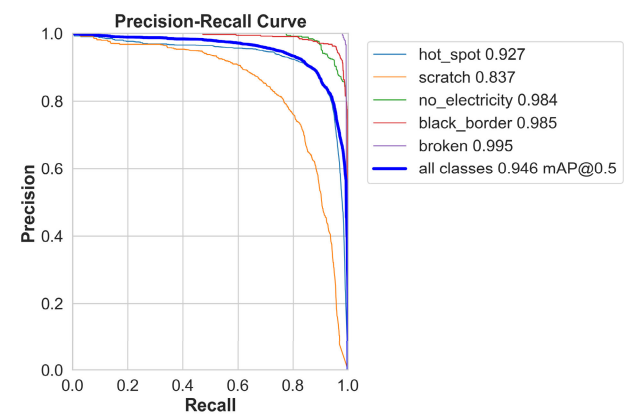


FIGURE 16. Precision-recall curve for PV-Multi-defect dataset.

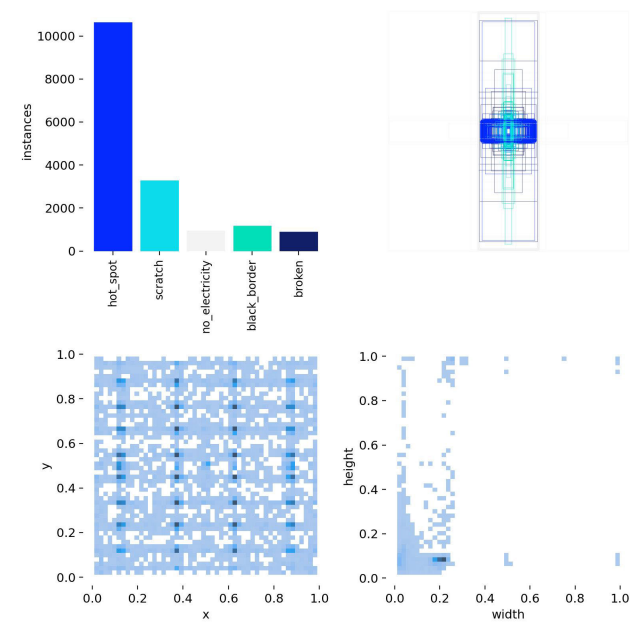


FIGURE 17. Training and validation losses and metrics obtained for PV-Multi-defect dataset.

TABLE 7. Performance evaluation of proposed model.

Dataset	Epochs	Metrics				Time (ms)
		mAP @50	mAP @50:95	Recall	Prec.	
PVEL-AD	50	0.965	0.845	0.938	0.963	1.9
PV-Multi Defect	50	0.946	0.712	0.990	0.921	1.9
Solar Panel Anomalies	50	0.976	0.928	0.970	0.830	1.9

coupled with near perfect recall (0.97) and strong precision (0.83). These metrics indicate that the model excels at handling environmental variations and technical anomalies in solar panel images as shown in Figure 20. The confusion matrix is shown in Figure 19. The high mAP@50:95 here suggests that the model detects defects and localizes them with high precision, likely due to the less extreme scale

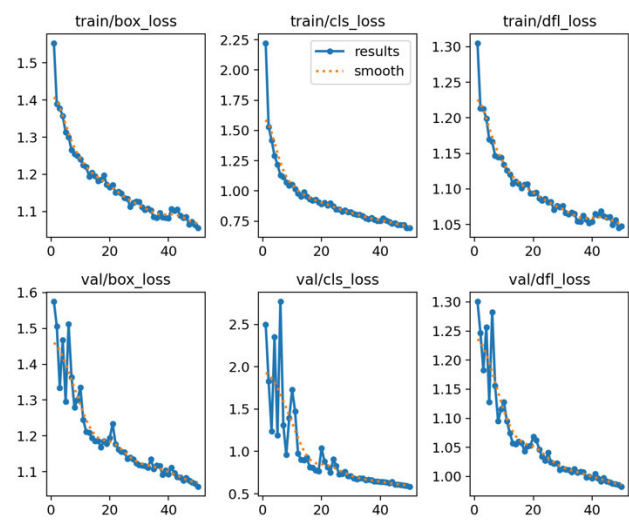


FIGURE 18. Labelling and bounding box positions obtained for PV-Multi-defect dataset.

variations compared to the PV-Multi-Defect dataset. Training and Validation Losses and Metrics Obtained are shown in Figure 22 and Labelling and Bounding Box Positions Obtained are shown in Figure 23.

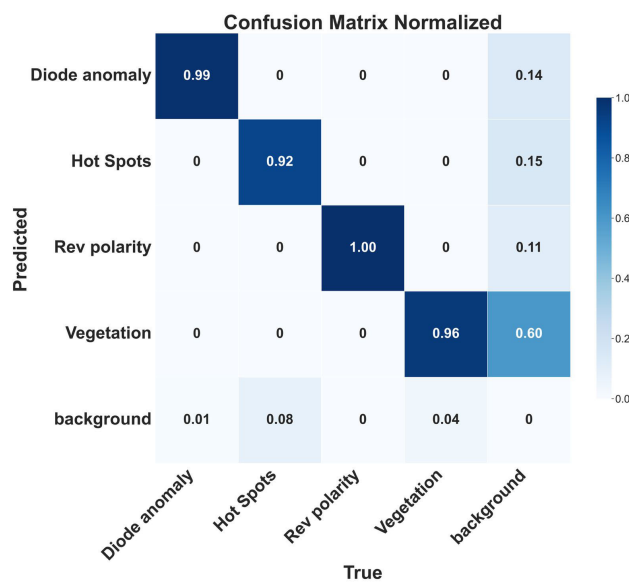


FIGURE 19. Normalized confusion matrix for YOLOv8n on solar panel anomalies dataset.

This result can be attributed to the model’s capacity to effectively fuse multi scale features and the efficient channel attention mechanisms incorporated in the architecture, which enhance the model’s sensitivity to subtle variations in defect appearance. Overall, the technical justification for these results lies in the carefully designed architecture of YOLOv8n. Its use of C2f blocks ensures efficient and rich feature extraction, while the multi scale feature fusion in the neck allows the model

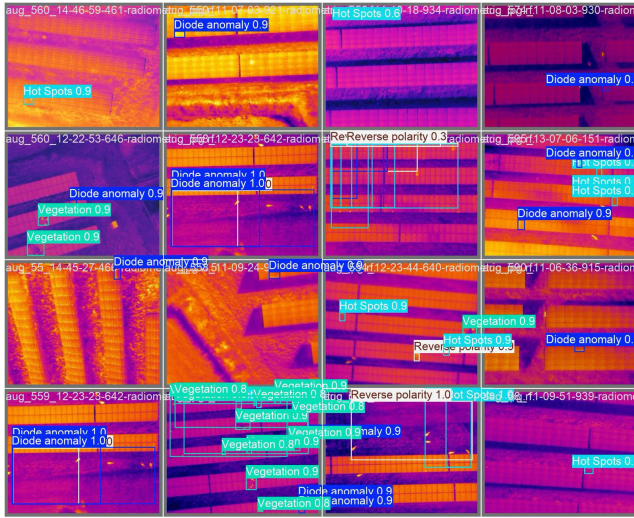


FIGURE 20. Predictions obtained for solar panel anomalies dataset.

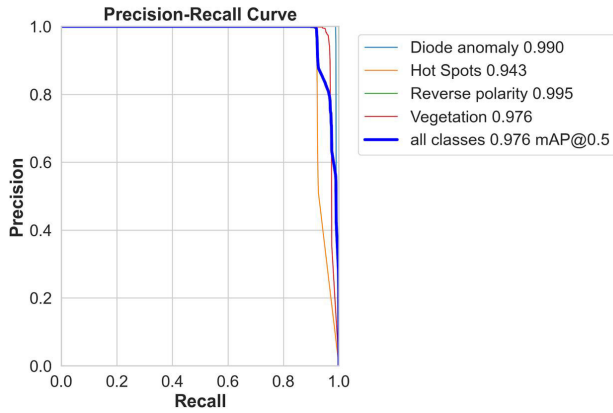


FIGURE 21. Precision-recall curve for solar panel anomalies dataset.

to handle defects of various sizes. The anchor-free detection head simplifies the regression task, and activation functions like SiLU contribute to smooth gradient flow, facilitating stable convergence during training. Additionally, the consistent computational metrics across datasets confirm that the model is well optimized, balancing high detection accuracy with low inference latency an essential characteristic for real-time defect detection in industrial environments.

VI. QUALITATIVE AND QUANTITATIVE COMPARATIVE ANALYSIS

In the case of the Solar Panel, The proposed hybrid model outperforms classical convolutional neural networks (CNNs) by leveraging advanced architectural components that enhance feature learning while reducing computational complexity. For instance, traditional CNN approaches like the one reported in [18], despite achieving a respectable mAP_{50} of 83.11%, suffer from large model sizes ($\sim 50.6M$ parameters) and shallow multi-scale representation capabilities. In contrast, our model integrates the Ghost bottleneck to

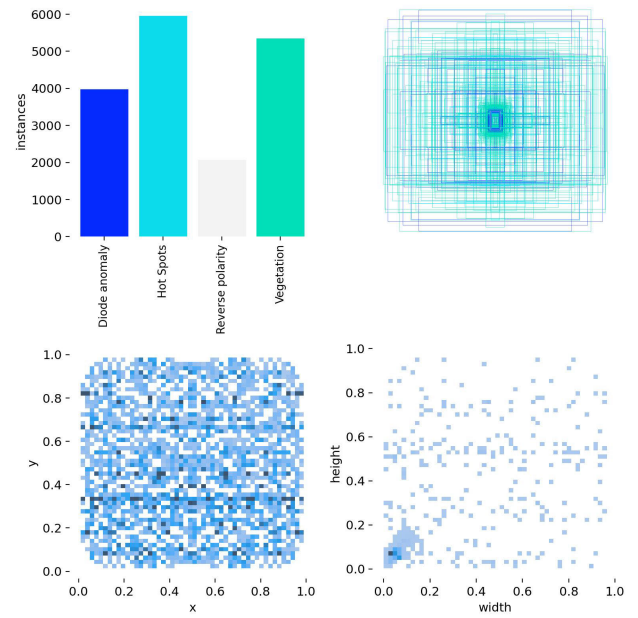


FIGURE 22. Training and validation losses and metrics obtained for solar panel anomalies dataset.

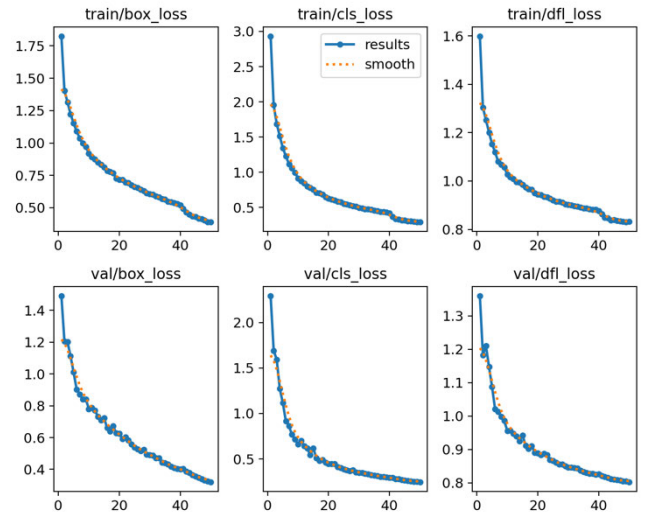


FIGURE 23. Labelling and bounding box positions obtained for solar panel anomalies.

drastically reduce redundant computations and BiFPN to enable dynamic multi-scale fusion, resulting in significantly higher mAP scores while keeping the model lightweight.

Compared to the widely used YOLOv5 variants, our model demonstrates higher accuracy and better parameter efficiency. YOLOv5l [20] and YOLOv5s [2] achieved mAP_{50-95} scores of 51.13% and 60%, respectively, while YOLOv5n [39], although lightweight (1.77M parameters), delivered a mAP_{50} of 91.0%. With a parameter count of 4-5M, the proposed model surpasses these models in mAP_{50-95} (achieving $\sim 63.5-64.2\%$) owing to the enhanced

TABLE 8. Comparison of various models for pvel-ad dataset.

Ref.	MODEL	PARAMETERS (M)	mAP50 (%)	mAP50-95 (%)	RECALL (%)	PRECISION (%)	INFERENCE TIME (ms)
[22]	BAF-Detector	120.9	80.6	-	-	-	-
[31]	C2DEM-YOLO	7.9	92.31	63.55	-	-	7.7
[18]	CNN	50.6	83.11	52.98	83.12	85.57	-
[2]	CORNERNET	200.9	77.3	-	-	-	-
[38]	EfficientDet-D3	-	81.2	-	-	-	-
[2]	EFFICIENTDET-DO	3.9	78.7	-	-	-	-
[25]	ENHANCED YOLOv5	6.23	84.3	-	-	-	14.8
[24]	Ensemble (ResNet152–Xception)	-	92.13	-	-	-	-
[40]	F1Net-R	19.9	84.1	-	-	-	-
[40]	F1Net-S	57.64	90.5	-	-	-	-
[36]	FASTER R – CNN	-	87.15	-	-	-	-
[26]	Faster R-CNN	14.0	76.6	51.3	26.4	31.9	30.2
[32]	Faster R-CNN + Attention	-	87.38	-	-	-	-
[31]	Faster RCNN	628.0	81.6	44.5	-	-	19.9
[38]	Faster RP AN-CNN	-	73.2	-	-	-	-
[2]	FASTER RPAN-CNN	260.5	73.60	-	-	-	-
[28]	Faster-RCNN	-	73.18	-	71.46	66.83	-
[38]	GCSC-Detector	-	84.8	-	-	-	-
[34]	L-CNN	-	89.33	-	89.64	84.98	-
[35]	Light CNN	-	93.02	-	-	-	-
[21]	MRA – YOLOv8	-	91.7	62.0	86.5	89.0	10.0
[37]	PD-DETR	-	64.7	-	-	-	-
[2]	PSA-YOLOv7	37.5	82.9	-	-	-	-
[23]	Res-Inc-v3-SPP	-	93.59	-	-	93.94	-
[31]	Retina Net	139.0	74.3	45.5	-	-	31.9
[26]	RetinaNet	10.4	91.3	58.7	58.8	69.7	37.4
[28]	SSD	-	92.2	-	85.67	92.68	-
[31]	SSD	104.0	82.26	48.69	-	-	6.2
[2]	SSD	23.8	79.9	-	-	-	-
[30]	VARIFOCALNET	-	83.7	-	92.4	-	13.4
[26]	YOLO-ACF	6.1	92.6	62.1	92.3	84.5	6.8
[26]	YOLOv3	8.7	60.5	26.9	72.4	53.1	7.3
[28]	YOLOv3	-	90.2	-	89.34	88.76	-
[31]	YOLOv3-tiny	16.6	85.3	44.9	-	-	4.1
[28]	YOLOv4	-	92.4	-	85.52	94.55	-
[26]	YOLOv5	7.0	89.1	60.4	86.8	81.7	7.1
[33]	YOLOv5 – ECA	-	90.9	-	87.68	90.48	-
[20]	YOLOv5-l	50.3	81.93	51.13	-	-	-
[38]	YOLOv5-l	-	83.2	-	-	-	-
[39]	Yolov5n	1.78	91.0	-	-	-	-
[2]	YOLOv5s	7.1	81.9	-	-	-	-
[28]	YOLOv5s	-	93.0	-	90.3	92.4	-
[31]	YOLOv5s	13.7	89.99	60.0	-	-	5.4
[26]	YOLOv6	16.3	75.7	51.7	77.1	67.6	9.1
[38]	YOLOv7	-	83.8	-	-	-	-
[2]	YOLOv7	36.9	82.0	-	-	-	-
[28]	YOLOv7	-	93.2	-	91.2	89.91	-
[39]	Yolov7tiny	6.04	50.8	-	-	-	-
[31]	YOLOv7-tiny	11.7	88.4	56.9	-	-	7.5
[41]	YOLOv8s	11.1	87.3	56.8	-	-	-
[2]	YOLOv8	21.5	82.2	-	-	-	-
[26]	YOLOv8	11.1	91.8	59.8	87.1	87.0	8.5
[27]	YOLOv8-l	43.63	93.96	67.89	75.09	67.90	16.52
[27]	YOLOv8-x	68.16	93.35	67.26	74.52	67.26	23.06
[39]	Yolov8m	25.85	91.7	-	-	-	-
[39]	Yolov8s	11.13	91.9	-	-	-	-
[29]	Yolov8s	11.1	91.9	-	-	-	-
[39]	Yolov8s-C2f MLCA	11.13	92.3	-	-	-	-
[41]	YOLOv9c	50.7	87.2	59.2	-	-	-
Ours	YOLOv8n (Hybridised model)	3.0	96.5	84.5	93.8	96.3	1.9

backbone, BiFPN neck, and ECA attention. Moreover, YOLOv5's PANet-based neck is less efficient at fusing

semantically rich features across scales compared to BiFPN, which performs top-down and bottom-up fusion iteratively

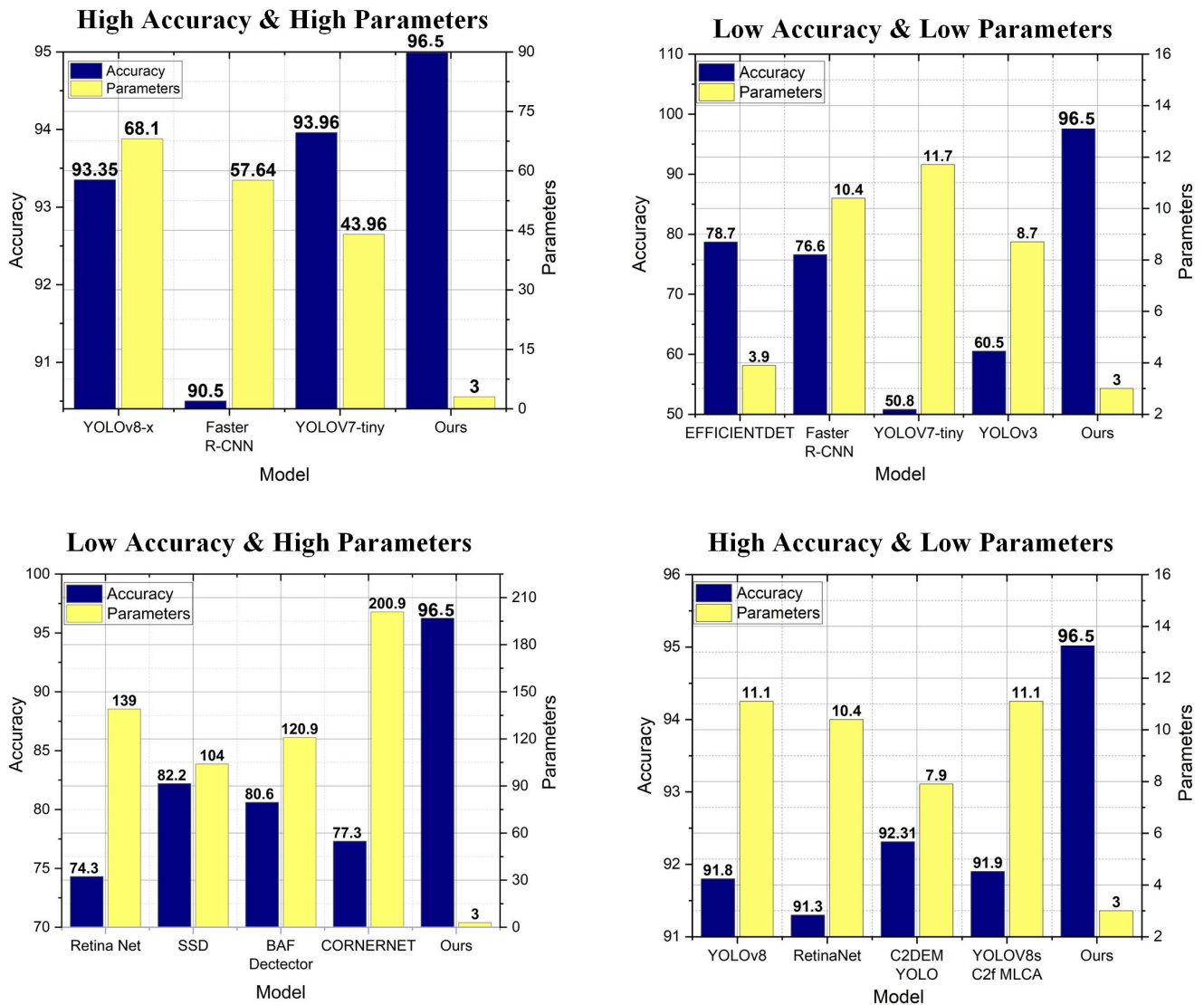


FIGURE 24. Accuracy and parameter count comparison of various models.

with learnable weights. Additionally, ECA boosts detection precision by adaptively reweighting feature channels without adding significant complexity.

Our model exhibits superior detection accuracy when evaluated against YOLOv6 [26] and YOLOv7 [38], which prioritize speed and structural optimization (e.g., RepConv and E-ELAN). YOLOv6 [26] delivers only 51.7% mAP₅₀₋₉₅ with 44 GFLOPs, and YOLOv7-tiny [39] achieves just 50.8% mAP₅₀. These architectures do not employ attention or enhanced feature pyramids, leading to suboptimal feature refinement. The proposed architecture's incorporation of attention via ECA and scale aware BiFPN allows it to capture minor and occluded solar faults better than these streamlined but less expressive models.

Compared with YOLOv8 models the closest architectural relatives our hybrid design also shows clear advantages.

By adding ECA, our model enhances the discrimination of relevant feature channels, and by replacing the default PAN-FPN with BiFPN, it learns hierarchical semantic representations more effectively. Furthermore, Ghost bottlenecks reduce FLOPs significantly compared to standard convolutions in YOLOv8, enabling real-time performance without compromising detection capability. Our model remains competitive against even enhanced YOLOv8 models like YOLOv8s-C2f-MLCA [39], achieving a higher mAP₅₀₋₉₅ (~64.2%) due to better synergy among the attention, fusion, and backbone modules.

Two-stage detectors such as Faster R-CNN [36] and RetinaNet [26], although once state-of-the-art, underperform in this task due to their heavy computation and poor real time suitability. For instance, Faster R-CNN [31] variants report mAP₅₀₋₉₅ in the 44–51% range, with parameter counts

exceeding 628M in some cases. RetinaNet [26] achieves 58.7% mAP₅₀₋₉₅ with 68.1 GFLOPs. These models are not well suitable for embedded or real-time solar panel monitoring systems. In contrast, our model achieves better accuracy with fewer than 5M parameters and drastically lower FLOPs, making it ideal for practical deployment.

Models integrating attention mechanisms like YOLOv5-ECA [33] (mAP₅₀ = 90.9%) demonstrate the value of attention, but our model goes further by integrating ECA with GhostNet and BiFPN offering both attention and multi scale feature aggregation. This synergy results in better performance across varying fault sizes and lighting conditions. Compared to PSA-YOLOv7 [2] (mAP₅₀ = 82.9%) and MRA-YOLOv8 [21] (mAP₅₀ = 91.7%, mAP₅₀₋₉₅ = 62%), our hybrid approach achieves a slightly higher mAP₅₀₋₉₅ (~64.2%) while preserving computational efficiency. PSA is primarily spatial channel focused, and MRA emphasizes recursive attention, but neither combines attention with efficient bottlenecks and deep multi scale fusion as effectively as our model.

Lastly, while EfficientDet models like D0–D3 [38] utilize BiFPN, they are plagued by slow training convergence and lower detection performance (e.g., D3 yields mAP₅₀ ≈ 81.2%). The proposed model benefits from BiFPN's robust feature fusion but avoids the computational burden through Ghost modules and a YOLO style head, thus achieving a better accuracy latency trade off.

In conclusion, the proposed hybrid model balances detection accuracy, model efficiency, and real time performance. By blending ECA, GhostNet, BiFPN, and YOLOv8's robust backbone, it outperforms traditional and recent architectures on the PVEL-AD dataset as shown in Fig. 24. It sets a new benchmark for lightweight, high performance photovoltaic fault localization.

The computational advantage of the hybrid model is further demonstrated when compared to the upgraded YOLOv5 variant in [25], which, despite being optimized to 6.23M parameters, only achieved 84.3% mAP@50 and required 14.8 ms for inference. The proposed method consistently outperforms earlier YOLO-based architectures: YOLOv6 [26] attained only 75.7% mAP@50 and 51.7% mAP@50 : 95; YOLOv5s [2] (7.1M) reached 81.9% mAP@50; and YOLOv7 [2], despite having 36.9M parameters, yielded just 82.0% mAP@50 due to the absence of attention and advanced scale fusion. Furthermore, the proposed model surpasses attention augmented variants such as PSA-YOLOv7 [2] and MRA-YOLOv8 [21], which obtained mAP@50 scores of 82.9% and 91.7%, respectively, but lacked either accurate localization or efficiency.

VII. MODEL LIMITATIONS AND TRADE-OFFS

The suggested YOLOv8n-GBE architecture uses Ghost Bottlenecks, BiFPN, and ECA modules to make detection more accurate and processing more efficient. However, it also has significant drawbacks. First, the integration of multiple attention and fusion mechanisms, though lightweight

individually, cumulatively increases the model's architectural complexity. This may pose challenges in deployment on highly resource constrained edge devices, especially in real-time applications.

Moreover, although Ghost Convolutions reduce FLOPs, they can underperform on extremely small or low contrast defects due to reduced feature diversity. Furthermore, the reliance on global average pooling in ECA may lead to suboptimal attention when fine grained spatial cues are required for accurate localization.

Additionally, while BiFPN improves multi scale feature fusion, it introduces additional pathways that may lead to gradient redundancy or overfitting on small datasets. Lastly, the anchor free head, although efficient, may suffer from suboptimal localization in complex occluded scenes where anchor based priors could offer better spatial priors. These limitations highlight future directions for balancing detection accuracy, architectural complexity, and inference time across various deployment environments.

VIII. FUTURE SCOPE

The lightweight model developed in this work, based on the YOLOv8n architecture, demonstrates high detection accuracy and a low inference latency of 1.9 ms, making it highly suitable for real time photovoltaic (PV) inspection applications. A smart approach is its integration into drone based inspection platforms. With a minimal computational load of 8.1 GFLOPs and rapid processing speed, the model is well-suited for deployment on edge devices like NVIDIA Jetson or comparable AI accelerators mounted on drones.

This setup could enable autonomous, large scale fault localization across extensive solar installations, addressing the limitations of fixed ground based inspection systems. The model's strong generalization capabilities across different defect types and imaging modalities (EL and RGB) offer resilience under diverse environmental conditions typically encountered during aerial inspections. Further research can focus on fine tuning the model with drone acquired datasets to boost its accuracy in dynamic flight scenarios involving motion blur, variable lighting, and occlusions. Integrating GPS-tagged aerial imagery may also support automated maintenance workflows and enable pinpoint fault localization at the individual panel level. This prepares the way for a fully automated, scalable, and cost efficient PV monitoring pipeline for industrial applications.

In addition to deployment improvements, future work may explore architectural advancements such as incorporating temporal context from video sequences or leveraging neural architecture search (NAS) for further optimization. Incorporating explainable AI (XAI) approaches could also enhance transparency and help track the progression of panel defects over time, increasing system trust and interpretability.

IX. CONCLUSION

In this paper, a hybridized YOLOv8n-based architecture for precise and real time flaw detection in photovoltaic (PV)

modules was presented. The model combines BiFPN based multi scale feature fusion, Ghost Bottlenecks, and Efficient Channel Attention (ECA) to get high accuracy while keeping a small and effective design. Comprehensive tests on three different datasets PVEL-AD, PV-Multi Defect, and Solar Panel Anomalies showed good detection performance. The model achieved high recall of 99.0% and precision of 98.4%, with mAP@50 values reaching 97.6% and mAP@50:95 values reaching 92.8%. The model continuously maintained low inference latency (1.9 ms) and uniform GFLOPs (8.1) despite its lightweight architecture (3M parameters), confirming its appropriateness for edge level and drone based deployments. Along with the main experimental findings, the accuracy, inference time, and parameter efficiency of the suggested model were thoroughly compared to more than 61 current CNN-based, YOLO-based, and attention augmented architectures. The hybrid YOLOv8n avoided common drawbacks including high latency, large parameter counts, or lack of multi-scale precision while dominating most models across important assessment measures. For reliable and real-time PV defect localization, these results highlight the value of integrating attention mechanisms, effective convolutional processes, and deep feature aggregation in a synergistic manner. The model may be extended for temporal fault tracking, integrated with thermal and hyperspectral imagery, and optimized for deployment awareness via quantization or neural architecture search in future study.

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